

TEST 1

ANSWER ALL QUESTIONS

DURATION: 1 HOUR

Write your name, IC No. and section on every page of your answer sheets.

Scan all your answers and upload the file in .pdf via Blackboard. Make sure you upload the correct file.

QUESTION 1 (10 MARKS)

- a) i. What happen to a neutral metal rod when a positively charged metal plate is brought to contact?

Neutral metal rod will be charged by conduction method

(1 mark)

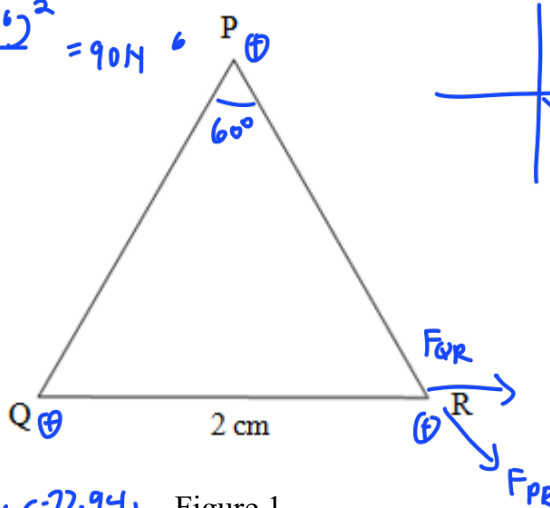
if in contact $\Rightarrow$ conduction $\Rightarrow$ acquires same chargeif no contact $\Rightarrow$ induction $\Rightarrow$ metal rod neutral but has separate charge

- ii. Three point charges are fixed at the vertices of an equilateral triangle PQR as shown in Figure 1. Determine the magnitude and direction of resultant electric force on charge at point R due to charges at point P and Q. Let charges at each point is $+q = 2.0 \mu\text{C}$. $F = 155.7 \text{ N}$, -30°

$$F_{PR} = \frac{kQq}{r^2} = \frac{9 \times 10^9 (2.0 \times 10^{-6})^2}{(0.02)^2} = 90 \text{ N}$$

$$F_{QR} = -90 \text{ N}$$

	x	y
F_{PR}	$90 \cos 60^\circ$	$-90 \sin 60^\circ$
F_{QR}	90	0
	135	-77.94



$$F = \sqrt{135^2 + 77.94^2}$$

$$= 155.9 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{-77.94}{135} \right)$$

$$= -30^\circ$$

Figure 1

(4 marks)

- b) Figure 2 shows two charges of $+2q$ and $-q$ separated by a distance of $2r$. Point X is located mid-point between the two charges.

- i. Deduce the expression for electric field at point X. $E = 3kq/r^2$

$$E_p = E_1 + E_2 = \frac{kQ}{r^2} + \frac{kQ}{r^2} = \frac{k(2q)}{r^2} + \frac{k(q)}{r^2} = \frac{2kq + kq}{r^2} = \frac{3kq}{r^2} \quad (3 \text{ marks})$$

- ii. If $q = 2\mu\text{C}$ and $r = 0.6 \text{ cm}$, determine the strength of electric field at point X and its direction. $1.5 \times 10^9 \text{ N/C}$, to the right

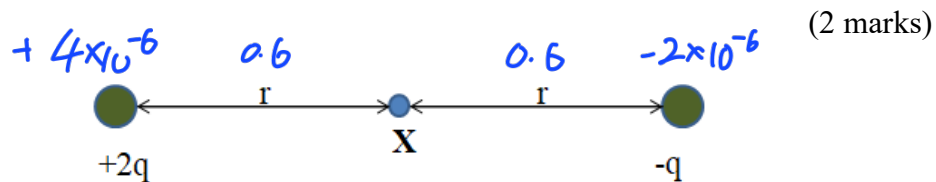


Figure 2

$$E = \frac{3(9 \times 10^9)(2 \times 10^{-6})}{(0.6 \times 10^{-2})^2} = \frac{F}{q_0} = \text{N/C}^{-1}$$

$$= 1.5 \times 10^9 \text{ N/C}$$

QUESTION 2 (10 MARKS)

- a) If a negative charge is initially at rest in an electric field, will it move toward a region of higher potential or lower potential? And what about the positive charge? How in each case does the potential energy of the charge change?

The negative charge will move toward a region of higher potential and the positive charge will move toward a region of lower potential (2 marks)

In both cases, the potential energy of the charge will decrease

- b) A lightning flash transfers 4.0 C of charge and 4.8 MJ of energy to the Earth. Between what potential difference did it travel? $1.2 \times 10^6 \text{ V}$

$$q = 4 \text{ C}, E = 4.8 \times 10^6 \text{ J} \quad V = \frac{W}{q} = \frac{4.8 \times 10^6}{4} = 1.2 \times 10^6 \text{ V} \quad (2 \text{ marks})$$

- c) Figure 3 shows two charges (Q_1 and Q_2) and two points (A and B). Calculate:

- i. Electric potential at point A and point B. $V_a = -449.5 \text{ V}$, $V_b = -316.7 \text{ V}$

$$V_A = \frac{9 \times 10^9 (3 \times 10^{-7})}{0.09} + \frac{9 \times 10^9 (-5 \times 10^{-7})}{0.06} = -450 \text{ V} \quad V_B = \frac{9 \times 10^9 (3 \times 10^{-7})}{0.11} + \frac{9 \times 10^9 (-5 \times 10^{-7})}{0.08} = -317 \text{ V} \quad (3 \text{ marks})$$

- ii. Potential difference between A and B. 132.8 V

$$V_{AB} = V_A - V_B = -133 \text{ V} \quad (2 \text{ marks})$$

- iii. Work done to bring a charge $+1 \mu\text{C}$ from A to B. $1.33 \times 10^{-7} \text{ J}$

(1 mark)

$$V = \frac{W}{q_0}, \quad W = V q_0 = -133(1 \times 10^{-6}) = -1.33 \times 10^{-4} \text{ J}$$

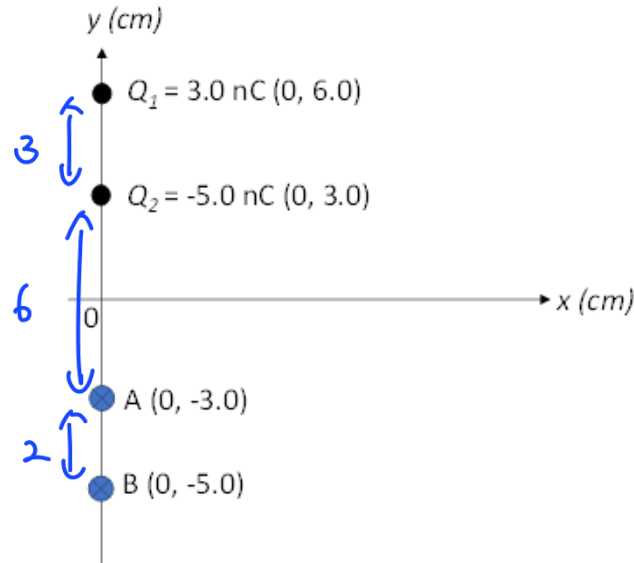


Figure 3

QUESTION 3 (10 MARKS)

- a) A parallel plate capacitor has two circular plates of diameter 20.0 cm with air gap between them. The plates are separated by distance of 0.5 cm. Calculate:

- i. The capacitance of the capacitor. $5.56 \times 10^{-11} \text{ F}$

$$A = \pi r^2 = 0.03142 \text{ m}^2 \quad d = 0.5 \times 10^{-2} \text{ m} \quad C = \frac{A \epsilon_0}{d} = \frac{0.03142 (8.85 \times 10^{-12})}{0.5 \times 10^{-2}} = 5.56 \times 10^{-11} \text{ F} \quad (3 \text{ marks})$$

- ii. The charge stored in the capacitor if the plates are connected to a battery of 1.5 V. $8.34 \times 10^{-11} \text{ C}$

$$Q = ?, \quad V = 1.5 \text{ V}$$

(1 mark)

$$Q = CV = 5.56 \times 10^{-11} (1.5) = 8.34 \times 10^{-11} \text{ C}$$

- b) Three identical capacitors, each of capacitance $6 \mu\text{F}$ are connected to a 12 V battery as shown in Figure 4. Calculate:

- i. The equivalent capacitance. $4 \times 10^{-6} \text{ F}$

$$C_{XY} = C_X + C_Y = 12 \mu, \quad \frac{1}{C_{\text{eff}}} = \frac{1}{C_2} + \frac{1}{C_{XY}} = \frac{1}{6} + \frac{1}{12}, \quad C_{\text{eff}} = 4 \mu \text{F} \quad (2 \text{ marks})$$

- ii. The potential difference across capacitor Z. 8 V

$$Q_T = V_{\text{eff}} C_{\text{eff}} \quad Q_T = Q_2 = Q_{XY} = 48 \mu \quad Q_Z = C_Z V_Z \quad (1 \text{ mark})$$

$$= 12(4 \mu) \quad 48 \mu = 6 \mu(V_Z) \quad V_Z = 8 \text{ volt}$$

$$= 48 \mu$$

- iii. The charge on each capacitor. $Q_Z = 48 \times 10^{-6} \text{ C}$, $Q_X = Q_Y = 24 \times 10^{-6} \text{ C}$

$$Q_Z = 48 \mu \text{ C}$$

$$Q_X = V_X C_X \quad (3 \text{ marks})$$

$$= 4 \mu \times 6 \mu$$

$$= 24 \mu \text{ C}$$

$$V = V_Z + V_{XY}$$

$$12 = 8 + V_{XY}$$

$$V_{XY} = 4 \text{ V}$$

$$Q_Y = V_Y C_Y$$

$$= 4 \mu \times 6 \mu$$

$$= 24 \mu \text{ C}$$

In series

$$V_{XY} = V_X = V_Y$$

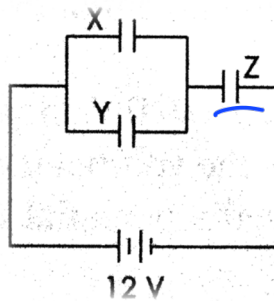


Figure 4